NOVAKID · LEAD, STUDENT EXPERIENCE SQUAD · CANDIDATE TASK

One learner, three products: connecting what Novakid already knows about a child

November 2026

CLICKABLE PROTOTYPE

sergeyhakobyan1992.github.io/novakid-my-album-prototype

A copy of Kids Area with the new feature. Laptop or tablet, landscape, sound on.
Guide with screenshots: slide 8.

CONTENTS

Part 1 Strategy · slides 2–5
Part 2 Kids Area initiative · slides 6–7
Prototype guide · slide 8
Part 3 Content · 9 · Part 4 · 10 · Sources · 11

Mia (8) and her teacher Ms. Anna are fictional. Novakid facts come from the parent and child accounts; numbers marked “assumption” are mine.

One week of Mia: each product starts from zero

Level
One of 7 stages, Pre-K to Level 6. Mia is in Level 1

→

Unit
A topic and its language. Level 1 has 13; hers is Unit 6 Food

→

Lesson
Unit 6 has four lessons, then a Unit Review

→

Competency
A skill: “speak about food using like / don’t like”

The competency model already exists (parent account, Achievements), but it only says “unlocked”: no mastery, no evidence, no source. The Learning Engine adds those three.

Mon · 1-1 lesson

Says “I no like peas”, mixes up peas and grapes. Nothing recorded.

1. Errors stay in the lesson

Tue · Kids Area

Daily quest “play any game”: Memo Mix with *family, bike*, then 16 min of Streams.

2. Practice ignores her unit

Wed · Homework

Listen and pick: she recognises the words, but never says them.

3. Trains recognition only

Thu · Group class

Topic: weekends. The teacher doesn’t know her unit; Mia stays silent.

4. No shared profile

Fri · Pandy AI

Chats about Minecraft, half in her native language.

5. Doesn’t know the week’s goal

Sat · Parent

68 competencies “unlocked”, one listed four times.

6. The payer sees no progress

Mon · 1-1 lesson

Can’t tell if she practised: 5–7 min of re-checking.

7. Practice never reaches the teacher

Hurts learning most: 1 and 7. The lesson-to-practice loop is cut.

Hurts retention most: 6, then 7. Parents and teachers see nothing.

Source: Novakid child and parent accounts; Mia’s week is a composite. · 2

Ten signals on one five-step ladder per competency

A **signal** is a child's answer to a task tagged with a competency ID from Novakid's existing competency model (Achievements). Every attempt, hint and response time is logged. Time spent and attendance are context, not evidence. A step needs its own kind of evidence: recognition games never lift a word past step 2.

1 · Met in lesson	2 · Recognises	3 · Says it	4 · Remembers	5 · Uses it
Lesson record: unit and competencies taught 1-1 · system data. Context for every signal, not evidence	**#1 Audio → picture, first try** Kids Area · **medium**: 1 in 4 is a guess	**#4 Builds a phrase or spells a word** Kids Area · **medium**: options can be brute-forced	**#8 Correct 3+ days after the last contact** Kids Area, 1-1 · **reliable**: knowledge, not short-term memory	**#6 Teacher: can't / with help / alone** 1-1 · **noisy** (human), but the most valid speech signal
	#7 In-lesson interactives 1-1 · **medium**: the teacher may prompt	**#5 Says the structure in reply** AI Tutor · **noisy**: child speech recognition, LLM grading		**#9 Uses it in a new context** Group, AI Tutor · **noisy** and rare, strongest evidence

Across all steps

#2 Attempts and hints before the right answer
Kids Area · **reliable** if every attempt is logged

#3 Response time on known items
Kids Area · **noisy**: trend only

#10 Unit Review, competency checks
1-1 · **reliable**, once per chapter

One score per answer: correct × attempt factor (1 → 0.8 → 0.6…) × 0.6 if a hint was used, corrected for guessing. Average per competency → difficulty zone; 0.75–0.95 is the right challenge.

Based on: Abbakumov 2025 (signals, score, zones) · Shute 2011 (evidence in games) · Koedinger et al. 2012 (task → competency map) · Laufer & Goldstein 2004 (recognise before produce) · 3

The last lesson drives practice; practice comes back to the teacher

1-1 lesson writes
Unit and competencies taught. System data, zero teacher time

→

Learning Engine
Step per competency, next due date

→

Kids Area reads
Lesson words become album cards, played at the right step

→

Teacher card
Before the next lesson: “peas: remembers · don’t like: with help”

Why this loop first

- The data already exists and is reliable
- The reading side is all ours: about 6 weeks
- It feeds the teacher bond that drives retention
- Effect shows within weeks

---

1 · Practice habit
Kids Area active days (1.39 ↑) · W1 retention, levels 0–1 (>20%) · AI Tutor retention (30% → 36%)

2 · Practice quality
Tasks at the right difficulty (score 0.75–0.95) · first-try, no-hint accuracy

3 · Learning progress
Ladder steps gained per learner per week · 3-day recall · Unit Review results

4 · Lesson link
1-1 and group attendance · 1-1 lessons with practice in between

←

NORTH STAR

Weekly Learners with Retained Progress

Share of active learners who in a week moved 1+ target competency up a step and passed a check 3+ days later.

Counts knowledge, not stars; needs a return visit; moves weekly; parents get it.

←

Renewal
1-1 subscription; the parent decides

Guardrails: stars earned on mastered or too-easy content · teacher input ≤ 30 s per lesson · Pandy safety incidents · daily screen time

“Passed a check 3+ days later” follows the spacing and testing effects: Cepeda et al. 2006 · Roediger & Karpicke 2006 · 4

Measure first, close one loop, then widen it

	Q1 · weeks 1–13		Q2 · weeks 14–26	
Learning Engine 2 → 3 eng. (incl. data/ML)	Competency map, attempt log	Ladder statuses v1 (5 steps)	Teacher one-tap rating feeds steps	Pandy reads unit goals
Kids Area 3 → 2 eng., 1 designer	Baseline scores, too-easy list	Loop v0 + My Album A/B	Album v2: teacher piece, more units	Pandy practises album words
Teachers, parents, group 1 designer + partners		Teacher pre-lesson card, pilot	Parent progress page v2	Group class: transfer pilot
Content 1 eng. tooling, 4 method.	Tag tasks to competencies	Pipeline v1: spec → AI → checks	Every new unit: one spec for four products, soft launch to 5%	
Analytics shared analyst	Metric definitions, tracking	Experiments: My Album A/B, long-term holdout, content variant tests		

Not doing in these two quarters

- New game worlds: we re-point existing games
- More Streams; leaderboards
- Our own child speech recognition
- IRT, knowledge tracing before data allows
- Group changes beyond reading the profile
- New curriculum levels

Dependencies: new asks only in Q2, once value is shown

Curriculum	Sign-off on task-to-competency tagging
Classroom platform	Lesson-end rating: agree Q1, ship Q2
Teacher Operations	Co-own the pilot; cap of 30 s per lesson
Data engineering	Events into the existing progress service
Parent journey, Legal	Shared “report opened”; child-data review

Capacity: 6 engineers, 2 designers, a shared analyst, 4 methodologists. Bars show stages, not a committed plan. · 5

My Album: the last lesson becomes a collection

Why kids don't come back (hypotheses)

1. **Nothing pulls them back tomorrow.** Daily quests only count games, and kids wait weeks for the first shop item.
2. **Games ignore their lessons:** kids practise random words, not their unit.
3. **Nobody notices the effort:** teachers and parents never see it.
4. **Games ask for no English:** kids match shapes and passively watch Streams.

We target ages 7–9, levels 0–1, first 8 weeks. Kids form habits here, the W1 target sits here, and parents still doubt whether to renew.

We move active days per week first: the mechanic rewards only coming back on another day, and an A/B test shows it in 1–2 weeks.



A card = 5 puzzle pieces = the 5 steps from 1.2

1 Met in lesson · the teacher adds the word after the 1-1 lesson

2 Recognises · matches sound to picture (Memo Mix, Box Battle)

3 Says it · says or spells the word (Pinocchio, Bubbles, Racing, Snake)

4 Remembers · names the picture 3+ days later (Quiz Corner)

5 Uses it · says it to Ms. Anna, who taps ✓ → gold frame

Why they come back, and learn

- **Kids return to earn the next piece:** it unlocks after 3 days, and the button counts down. Spacing effect
- **Cards ask for a refresh:** they gather dust after 7, 14, 28 days. A miss drops a card to 3 pieces, but the child keeps it. Half-life regression
- **The card teaches:** the child taps it to hear the word and presses PLAY; the engine picks the game. Intrinsic integration
- **Stars reward progress:** kids earn +10 a piece, +5 a refresh, nothing for replays.
- **We skip streaks and random packs:** adults keep most long streaks, and the UK Children's Code warns against FOMO loops.
- **Adults join in:** the teacher asks for the word, the parent practises it at home. Competence, autonomy, relatedness

Cards go beyond words. A card can also teach a structure ("I don't like…") or a sound (l / r): the engine tracks competencies, not words.

We start with the simplest collectible. We build and test an album cheaply, then can grow it into a richer world, e.g. a Word Garden where lessons plant seeds.

More days, short sessions, and words that stay

Metric	Baseline	Expected change (A/B, segment)
Active days per week	1.39	Up: +15–20% in 4 weeks (assumption)
Time spent	—	Same or shorter sessions (3–6 min) on more days; Streams share down
Repeat use	—	More returns for the 3-day check; share of lesson cards reaching piece 4
W1 retention, levels 0–1	target >20%	Up, towards the target
AI Tutor retention	~30%	Up in Q2, once Pandy practises album words
Learning	—	North Star up; next-lesson accuracy higher on practised competencies

Parent guardrail: useful-time share

≥ 70% of Kids Area minutes go to current-unit tasks at the right difficulty. If active days rise while this falls, kids came for the album, not the learning.

MVP

Units 5–6 of Level 1; cards from the last lesson; pieces 1–4 via three existing games; the 3-day check. Teacher's piece and parent view: Q2.

Biggest assumption

"Next piece in 3 days" brings a 7–9-year-old back, rather than disappointing them.

Playtest
6–8 kids, 1–2 days → **Painted door**
in daily quest, 1 week → **A/B, 4 weeks**
+ long-term holdout

Stop if day-3 return isn't better after 2 weeks.

Reads from the engine: the last lesson's unit and competencies (exists today), each word's step and due date; the teacher's rating from Q2.

Sends back: every attempt, hint and time with its competency ID, 3-day checks, Pandy speech from Q2. This feeds the teacher card and the parent page.

If the engine isn't ready: cards follow the child's programme position; steps come from Kids Area's own logs. Only the teacher card and last piece wait.

Baselines from the task. Expected changes are hypotheses to test, not forecasts. · 7

PART 2 · PROTOTYPE GUIDE

Try it in five minutes

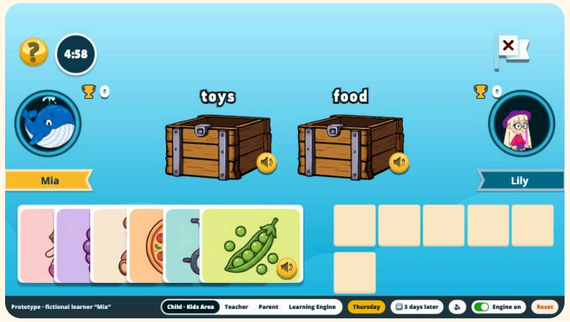
sergeyhakobyan1992.github.io/novakid-my-album-prototype · laptop or tablet, landscape, sound on. The bottom bar switches between the child, teacher, parent and engine views, jumps 3 days ahead and resets.



1 After the lesson: Ms. Anna added 6 Food words. Press OPEN.



2 A page per unit. Gold = all 5 pieces. Tap a card to hear it.



3 PLAY: the engine picks the game, here Box Battle.



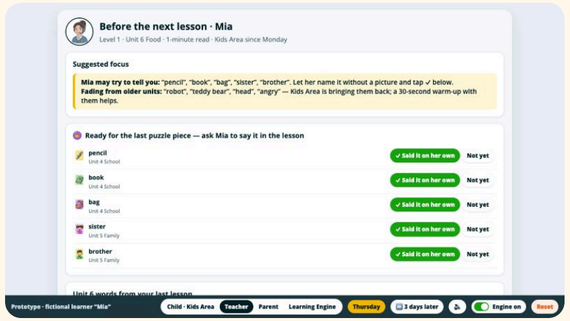
4 New pieces; stars only for progress.



5 “3 days later”: old cards got dusty and ask for a refresh.



6 Quiz Corner checks memory: first try, no hint.



7 Teacher: which words to ask for the last piece.



8 Parent: progress against the programme, words to practise.

Games and art are Kids Area’s own, rebuilt for the prototype; word pictures are generated. Fits 7+: voice-first, almost no reading, 3–5 minute sessions. · 8

One unit spec feeds four products; content is judged by recall

3.1 One unit, many products: Level 1, Unit 6 Food

Shared: the unit spec

- Competency IDs
- 11 food words; *I like / I don't like / Do you like...?*; sounds l, r
- Typical errors ("I no like") → distractors, hints
- Evidence rule per step
- Parent "can do" line
- One picture and audio library
- Safety: no diets, weight, "bad food"

→

1-1: four lessons teach the words; a warm-up re-checks them; the teacher rates "I don't like..." in one tap

Group: kids ask each other "Do you like...?" using the food words; one card pack per level band

Kids Area: words and pictures become album cards; "I no like" becomes a wrong option in games

AI Tutor: "Picnic with Pandy" asks about 3 foods, listens for *I like / don't like*, avoids diets

3.2 The pipeline

1 Unit spec	2 AI drafts	3 Pictures, audio	4 Auto checks	5 Review by risk	6 Safety QA	7 Soft launch
A methodologist, once per unit	Items, distractors, Pandy script, group cards	Library first; AI art in house style; voice	Level words, 1 right answer, grammar, assets	10% of approved templates; 100% of new ones	Classifier + human; child-like attacks on Pandy	5% of kids; difficulty from real answers

Bottleneck: step 5, four methodologists. More throughput, same quality: approve templates, not items · catch mechanical errors automatically · let soft-launch data flag weak items.

3.3 How we know content works

Axis	Works when
Engagement	Completion, replays ≥ Kids Area average; no drop-off cliff
Difficulty	Score 0.75–0.95; every distractor gets picked
Learning	3-day recall, next-lesson rating: practised beats unpractised
Safety	Zero Pandy incidents in zero-tolerance categories

Small segments: randomise within age (7–9, 10–12) × level (0, 1); variant A on some competencies, B on others in the same child. Primary: 3-day recall; 1–2 weeks.

Unit 6 Food, its lessons and competencies are taken from the real Level 1 programme. · 9

PART 4 · JUDGMENT SCENARIOS · EACH UNDER 150 WORDS

Weighing competing goals

A. Engagement without learning

I keep the habit, because it is the hard part, but I treat active days as a condition for learning, not the goal.

Week 1, diagnose. How much of the lift comes from children replaying level 0 games they have already passed? What is the difficulty score of those games (above 0.95 means they test nothing)? Can a noisy teacher rating move in six weeks at all?

Week 2, a fast A/B. Stars only for new progress, so replaying mastered content earns nothing, and daily quests point to the current unit. Primary metric: 3-day recall accuracy; guardrail: active days.

To leadership: “We built the habit, which is the hard part. Right now it points at stars, but parents pay for progress, and farming will cost their trust. In two weeks we change the reward rules. Success: active days drop no more than 10% while the learning metric rises.”

B. Foundation versus visible wins

It is a false choice if the engine is built in thin vertical slices rather than layers.

This quarter’s visible win uses data that already exists: the last lesson’s unit feeds My Album in Kids Area and a “can do” line for parents. The renewal campaign gets a real feature, and it is also engine v0: competency IDs, an attempt log, the first loop. The two engineers on the foundation are protected from extra feature requests.

To the commercial team: dated, concrete material for the campaign (“your child can now say…”, practice from their own lessons), and an honest list of what we won’t promise this quarter, such as full personalisation.

To engineering: the win is built on the future engine’s schema, not thrown away, and every stage ships something users see, so there are no silent quarters.

Both groups see one plan and a monthly demo.

C. Data at the cost of teacher time

Principle: ask teachers only for what we can’t get any other way.

1. **Automatic signals first:** in-lesson interactive results and a recall warm-up at the start of each lesson already cover recognition and memory.
2. **One tap, not a form:** the teacher rates 1–3 lesson competencies, pre-filled from the lesson plan, as can’t / with help / alone, on the end-of-lesson screen they already use. Under 30 seconds.
3. **Give time back:** a pre-lesson card on each student saves 5–7 minutes of re-checking, so teachers gain more than they give.
4. **A pilot with about 50 volunteers,** run with Teacher Operations: we measure input time, completion and agreement between raters on reference recordings. It becomes mandatory only after the benefit is proven.
5. **Later:** speech recognition on lesson recordings suggests the rating, and the teacher only confirms it.

Word counts: A 148 · B 147 · C 138 · 10

APPENDIX · SOURCES

What the decisions rest on

Measuring knowledge (Part 1)

- **Abbakumov, D. (2025).** *Psychometrics in EdTech: first steps.* MIF. → signals are observed answers; log every attempt; one score per answer; difficulty zones (0.75–0.95).
- **Shute, V. (2011).** Stealth assessment in computer-based games. → which game behaviour is evidence of which skill.
- **Koedinger, K., Corbett, A., Perfetti, C. (2012).** The Knowledge-Learning-Instruction framework. *Cognitive Science.* → task → competency map.
- **Laufer, B., Goldstein, Z. (2004).** Testing vocabulary knowledge. *Language Learning.* → recognition comes before production.
- **Corbett & Anderson (1995)** knowledge tracing; **Duolingo Birdbrain.** → the models we postpone until data can calibrate them.

Memory and motivation (Part 2)

- **Cepeda, N. et al. (2006).** Distributed practice in verbal recall tasks. *Psychological Bulletin.* **Roediger & Karpicke (2006).** Test-enhanced learning. → 3-day piece, Quiz Corner.
- **Settles, B., Meeder, B. (2016).** A trainable spaced repetition model for language learning. ACL. → cards fade after 7, 14, 28 days.
- **Habgood, J., Ainsworth, S. (2011).** Intrinsic integration in educational games. *J. of the Learning Sciences.* → the card is the content.
- **Ryan & Deci (2000)** self-determination theory; **Huang et al. (2020)** gamification meta-analysis, g ≈ 0.46. → competence, autonomy, relatedness.
- **Duolingo (2023)**, who keeps long streaks; **UK ICO** Children's Code guidance. → no streaks, no random packs.
- **Cambridge English** "can do" statements for parents. → the parent line.

Novakid facts (programme, competencies, Kids Area economy) come from the child and parent accounts, seen in November 2026. · 11